Describe It, Build It

Engineers and scientists have to explain their ideas so clearly that someone else can build them. In this event, one partner writes the directions and the other builds from them, with no pictures allowed.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 16	Write It, Build It: First Try
Lesson 2	Fri Nov 20	Name It and Place It
Lesson 3	Fri Jan 15	Straws, Cups, and Pipe Cleaners: Use the Plan
Lesson 4	Fri Feb 12	Mock Tournament: Write, Then Build
Lab I1	Before Lesson 3	The 6-Part Plan (home or an open Friday)
Lab I2	Before Lesson 3	Craft Words and Measuring (home or an open Friday)
Lab I3	Before Lesson 4	Check Sentences and Judging (home or an open Friday)

At the tournament

The Describer gets 25 minutes to write build directions for a mystery object using only words and numbers (3 sheets of wide-ruled paper); then the Builder, in a different room, has 20 minutes to rebuild it from those words, and the closest match wins.

Keep every handout for this event behind this divider, in order. Bring this binder to every Diagram & Decode Fridays meeting.

Describe It, Build It: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can explain how the event works: the Describer writes for 25 minutes, the Builder builds for 20 minutes in another room, and the closest match wins.	L1 L4	☐	☐
I can write using only words and numbers: no drawings, arrows, codes, or symbols, and punctuation only in its normal way.	L1 L2 I2	☐	☐
I can define every abbreviation at the top or the first time I use it (rt = right).	L2 I1	☐	☐
I can give each piece a full name: color, size, and type (red 2 by 4 LEGO brick).	L2	☐	☐
I can use clear position words (on top of, left end, centered, lined up with, hanging over) to say exactly where a piece goes.	L2 L3	☐	☐
I can start with an overview and a parts list before the step-by-step directions.	L3 I1	☐	☐
I can set the front of the object and describe which way each piece faces or points.	L3 I1	☐	☐
I can write steps in build order, bottom up, with one piece or connection per step.	L3 I1	☐	☐
I can use counting and measuring to describe size and distance (bumps, holes, centimeters, halfway).	L3 I2	☐	☐
I can describe objects made of craft materials like straws, cups, pipe cleaners, craft sticks, foam balls, and stickers.	L3 I2	☐	☐
As the Builder, I can read the whole description first, sort my parts, and follow one step at a time.	L2 L3 L4	☐	☐
I can add check sentences (Now your object is 3 layers tall) so my Builder can catch mistakes.	L4 I3	☐	☐
I can score a build like a judge: one point for each piece in the right place and connected the right way.	L1 L4 I3	☐	☐
I can manage time: the Describer uses all 25 minutes to write and reread, and the Builder works carefully but quickly (build time breaks ties).	L4 I3	☐	☐

Write It, Build It: First Try

TODAY I CAN

- I can explain how the event works: the Describer writes for 25 minutes, the Builder builds for 20 minutes in another room, and the closest match wins.
- I can write using only words and numbers: no drawings, arrows, codes, or symbols, and punctuation only in its normal way.
- I can score a build like a judge: one point for each piece in the right place and connected the right way.

One partner is the Describer and writes build directions. The other is the Builder and builds from the words alone. Today you try a mini version.

Today's steps: run it yourselves (22 minutes)

You need: Lesson 1 model bag (1 per pair, Describer only), Lesson 1 kit bag (1 per pair), File folder (1 per pair), Wide-ruled paper, Lesson 1 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Take turns reading The rules in one box out loud. Choose today's Describer; the other partner is the Builder (you switch next time). Stand the folder up between you.
2. Both 7 min. Set the timer for 7 minutes. Describer: open the model bag behind the folder and write build directions, words and numbers only. Builder: sit at the other end of the table, open the kit bag, and list every piece (color, size, type).
3. Both 5 min. Set the timer for 5 minutes. Describer puts the model back in its bag and hands over the directions. Builder builds from the words alone. Describer stays quiet: no hints, no pointing.
4. Both 4 min. Set the timer for 4 minutes. Take out the model and set it next to the build. Fill in Score our build: 1 point for each piece in the right place and connected the right way.
5. Both 3 min. Set the timer for 3 minutes. Answer Think about it together. Take the build apart into the kit bag, and put the model back in its bag, still built.

Check your work: Ask your Event Captain for the Lesson 1 answer card and check how you scored (a piece counts if it is in the right place, even if an earlier piece was wrong). If you disagree about a piece, the Event Captain decides.

The rules in one box

Describer	Sees the object and has 25 minutes to write directions. Gets 3 sheets of paper (front and back).
Builder	In a different room. Has 20 minutes to build from the directions. May use colored pencils (black, brown, green, blue, red).
Only words and numbers	No drawings, diagrams, arrows, or made-up symbols. Normal punctuation is fine.
Abbreviations	Allowed if you define them first: rt = right.
Scoring	Points for each piece in the right place and connected right. Breaking the writing rules drops you below every team that followed them.

Score our build

Piece (color and size)	Right place?	Connected right?	Point?
Total			/ 6

Think about it

1. Builder: which sentence confused you the most? Why?

2. Describer: what is one thing you would add to your directions next time?

At home before next time: Stack or line up 5 things from your kitchen (spoon, cup, box, and so on). In 10 minutes, write directions to rebuild it using only words and numbers. Take it apart and have a family member rebuild it from your words. Write down one word that confused them. Case File: write your 5 full piece names, your directions, and the score in Case #1: The Kitchen Stack (p. 7) for the Kitchen Describer badge.

Name It and Place It

TODAY I CAN

- I can write using only words and numbers: no drawings, arrows, codes, or symbols, and punctuation only in its normal way.
- I can define every abbreviation at the top or the first time I use it (rt = right).
- I can give each piece a full name: color, size, and type (red 2 by 4 LEGO brick).
- I can use clear position words (on top of, left end, centered, lined up with, hanging over) to say exactly where a piece goes.
- As the Builder, I can read the whole description first, sort my parts, and follow one step at a time.

Great directions name every piece clearly and say exactly where it goes. Today you practice both.

Today's steps: run it yourselves (22 minutes)

You need: Look-alike pile bag (12 pieces), Lesson 2 model bags (4, one per partner), Lesson 2 kit bags (4), File folders (2), Lesson 2 answer card. Set the timer for each step.

1. Both 3 min. Set the timer for 3 minutes. Check your Legal or not answers with the answer card. Remember: letters and numbers cannot be used as pictures.
2. Both 3 min. Set the timer for 3 minutes. Read the Part name recipe. Find a brick and a plate in the look-alike pile: a brick is tall, a plate is thin. Write sizes with 'by', like 2 by 4.
3. Both 6 min. Set the timer for 6 minutes. Parts Pick: one partner secretly picks 2 pieces from the pile and writes a full name for each. The other partner reads the names and picks the exact pieces. Switch.
4. Both 4 min. Set the timer for 4 minutes. Each partner takes a different model bag behind the folder and writes directions for it. Define abbreviations at the top and use words from Position words.
5. Both 4 min. Set the timer for 4 minutes. Give your partner your directions and the kit bag that matches your model. Read ALL the directions first, sort your parts, then build.
6. Both 2 min. Set the timer for 2 minutes. Take out the models and compare. Which position word helped most? Put builds back in the kit bags and models back in their bags.

Check your work: Put your build next to the model and count pieces in the right place and connected right. For each piece that is wrong, find the sentence that caused it and rewrite it on the back of your directions.

Legal or not legal?

1. Put the red brick -> on the blue brick.
◯ Legal ◯ Not legal

2. rt = right. Put the cup on the rt side of the stick.
◯ Legal ◯ Not legal

3. Stack 2 yellow bricks on the green plate.
◯ Legal ◯ Not legal

4. Bend the pipe cleaner into a V.
◯ Legal ◯ Not legal

5. Use the blue brick (the one with 8 bumps).
◯ Legal ◯ Not legal

6. Tape the straw @ the back of the cup.
◯ Legal ◯ Not legal

7. Put 2 stickers on the cup like a smiley face :)
◯ Legal ◯ Not legal

Part name recipe

Color	red, dark blue, light green (the judge will say if color does not matter)
Size	2 by 4 (2 rows of 4 bumps), 10 centimeters long
Type	brick (tall), plate (thin), straw, craft stick, cup

Special features	bendy end, one flat side, a hole in the middle

Position words

on top of **underneath** **left end** **right end** **front** **back** **centered** **lined up with**
hanging over by 1 bump **lying flat** **standing up** **upside down**
parallel (side by side, same direction) **perpendicular (crossing to make square corners)** **touching**
between **corner** **edge**

At home before next time: Do Independent Labs I1 (The 6-Part Plan) and I2 (Craft Words and Measuring) before Lesson 3. Also pick 3 things at home and say a full name for each out loud (color, size, type, special features). Case File: do Case #2: Name It and Place It (p. 8) for the Part Namer badge, and get all 7 Legal or not answers right (p. 4) for the Rule Keeper badge.

Straws, Cups, and Pipe Cleaners: Use the Plan

TODAY I CAN

- I can use clear position words (on top of, left end, centered, lined up with, hanging over) to say exactly where a piece goes.
- I can start with an overview and a parts list before the step-by-step directions.
- I can set the front of the object and describe which way each piece faces or points.
- I can write steps in build order, bottom up, with one piece or connection per step.
- I can use counting and measuring to describe size and distance (bumps, holes, centimeters, halfway).
- I can describe objects made of craft materials like straws, cups, pipe cleaners, craft sticks, foam balls, and stickers.
- As the Builder, I can read the whole description first, sort my parts, and follow one step at a time.

At the tournament the object might be made of craft materials instead of LEGO. Today you put your 6-part plan and your measuring skills together.

Today's steps: run it yourselves (27 minutes)

You need: Lesson 3 model bags (craft model, 1 per pair), Lesson 3 kit bags (1 per pair), Masking tape (2 rolls), File folders, notebook paper, Lesson 3 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Check your Do Now with the answer card. Take turns reading Craft words out loud. Shapes are described in words, not letters.
2. Both 10 min. Set the timer for 10 minutes. Switch roles from Lesson 2. Describer: write directions for the craft model behind the folder, using every box of The 6-part plan and measuring in centimeters with a ruler. Builder: on the Lab I1 Sample description, circle every number in red and box every step number in blue.
3. Both 8 min. Set the timer for 8 minutes. The Describer bags the model and hands over the directions. Builder: read everything, sort parts, then build with tape one step at a time, checking off each step with a colored pencil.
4. Both 5 min. Set the timer for 5 minutes. Take out the model and fill in Score our build. Answer Think about it. Take your build apart into the kit bag.

Check your work: Ask your Event Captain for the Lesson 3 answer card to check your score (the model has 5 pieces). For each piece that is off, find the measurement or word that caused it and fix it on your directions.

Craft words

Straw	long or short part, bendy part, open end
Pipe cleaner	wrap (go around), twist (turn two ends together), bend in half, bend to make a square corner
Cup	rim (top edge), bottom, upright or upside down, inside, outside
Craft stick	flat side, thin edge, rounded end
Measuring	centimeters from the end, halfway, one third of the way, touching, 2 finger widths is NOT exact

The 6-part plan (from Lab I1)

- [ ] Abbreviations defined at the top
- [ ] Overview: what it looks like, how big, how many pieces
- [ ] Parts list with full names
- [ ] Front: which way to turn the base
- [ ] Steps: bottom up, one piece per step, numbered
- [ ] Checks every few steps

Score our build

Piece	Right place?	Connected right?	Point?
Cup			
Craft stick			
Green pipe cleaner			
Straw			
Red pipe cleaner			
Total			/ 5

Think about it

1. Which measurement or craft word helped the Builder most?

At home before next time: Do Independent Lab I3 (Check Sentences and Judging) before Lesson 4. Then talk with your partner about who wants to be the Describer and who wants to be the Builder at the tournament. Case File: do Case #3: The Pipe Cleaner Caper (p. 9) for the Craft Measurer badge.

Mock Tournament: Write, Then Build

TODAY I CAN

- I can explain how the event works: the Describer writes for 25 minutes, the Builder builds for 20 minutes in another room, and the closest match wins.
- As the Builder, I can read the whole description first, sort my parts, and follow one step at a time.
- I can add check sentences (Now your object is 3 layers tall) so my Builder can catch mistakes.
- I can score a build like a judge: one point for each piece in the right place and connected the right way.
- I can manage time: the Describer uses all 25 minutes to write and reread, and the Builder works carefully but quickly (build time breaks ties).

This is a practice tournament with your real roles. Follow the rules exactly, like on tournament day.

Today's steps: run it yourselves (27 minutes)

You need: Lesson 4 model bags (new, sealed; Describers only), Lesson 4 kit bags, 3 sheets of wide-ruled paper per Describer, Rubric copies (2), File folders (2), Lesson 4 answer card. Set the timer for each step.

1. Event Judge 1 min. Read the rules out loud: only words and numbers, abbreviations defined, 3 sheets only. Describers take the model bags to one end of the room. Builders sit at a different table, facing away.
2. Event Judge 13 min. Set the timer for 13 minutes and say 'Go.' Call time left at 6, 3, and 1 minute. Describers write. Builders practice by reading and marking the Lab I1 Sample description.
3. Both 10 min. Set the timer for 10 minutes. Describers bag the models, hand over the papers, and stay silent. Builders open their kit bags and build. Write down each build time.
4. Both 3 min. Set the timer for 3 minutes. Take out the models and score with the rubric. Check the directions for any writing-rule problems. Fill in Our results and Look back.

Check your work: Go down the Describer checklist and Builder checklist on your handout and check each one you really did. Then read the Lesson 4 answer card to compare with what a strong description looks like.

Describer checklist before you hand it in

- [ ] Abbreviations are defined at the top or the first time I use them.
- [ ] Overview, parts list, and front are at the beginning.
- [ ] Steps are numbered, bottom up, one piece per step.
- [ ] I added check sentences.
- [ ] No drawings, arrows, codes, or letter shapes anywhere.
- [ ] I reread everything from the Builder's point of view.

Builder checklist

- [ ] I read the whole description first.
- [ ] I sorted my parts and matched the parts list.
- [ ] I built one step at a time and checked off each step.
- [ ] I compared my build to every check sentence.

Our results

Pieces correct	Pieces possible	Build time	Rule problems?

Look back

1. The first mistake in the build happened at step ___ because:

2. Our plan for tournament day:

At home before next time: Do one last practice at home: Describers write for 15 minutes about an object made of 8 to 10 pieces; Builders rebuild from the written directions. Put your final checklist in the front of this divider to read the morning of the tournament. Case File: record your practice in Case #4: The Final Test (p. 10) with 3 check sentences for the Check Writer badge, and read the Tournament day checklist (p. 14) that morning.

Lab I1: The 6-Part Plan

THIS LAB HELPS ME

- I can define every abbreviation at the top or the first time I use it (rt = right).
- I can start with an overview and a parts list before the step-by-step directions.
- I can set the front of the object and describe which way each piece faces or points.
- I can write steps in build order, bottom up, with one piece or connection per step.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Good directions follow a plan. Builders understand faster when every description has the same parts in the same order. In this lab you study a sample, then write your own.

Steps (28 minutes)

You need: Colored pencils (red, blue, green), 6 to 8 building pieces, Notebook paper. Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read The 6-part plan. Then find each of the 6 parts in the Sample description and number them 1 to 6 in the margin.
2. Me 5 min. Set the timer for 5 minutes. In the Sample description, underline every part name in red and every position word in blue. Circle the check sentences in green.
3. Me 3 min. Set the timer for 3 minutes. Build something with 6 to 8 pieces. Give it at least 2 layers. Decide which side is the front.
4. Me 10 min. Set the timer for 10 minutes. Write directions for your build on notebook paper using all 6 parts of the plan. Words and numbers only. One piece per step, bottom up.
5. Both 6 min. Set the timer for 6 minutes. Hide your build. Your Builder (a family member or partner) rebuilds it from your words while you stay quiet. Compare, count the pieces they got right, and answer Think about it.

Check your work: Look at the piece your Builder got wrong first. Find the sentence that caused it and rewrite it on the back of your directions. Then check your marked sample with the Lab I1 answer card in the Open Lab bin, or at the next Huddle.

The 6-part plan

1. Abbreviations: define them at the very top (br = brick).
2. Overview: what it looks like overall, how big, how many pieces.
3. Parts list: every piece with its full name and how many.
4. Front: tell the Builder how to turn the base so everyone agrees on front, back, left, and right.
5. Steps: bottom up, one piece per step, numbered.
6. Checks: every few steps, a sentence the Builder can use to check their work.

Sample description

Abbreviations: br = brick. Overview: This object is 3 layers tall and made of 4 LEGO pieces. Parts: 1 blue 2 by 4 br, 2 red 2 by 2 br, 1 yellow 2 by 4 br. Front: Lay the blue br flat with its long side facing you. That long side is the front. Step 1: Put 1 red br on the left end of the blue br, covering its 4 left bumps. Step 2: Put the other red br on the right end the same way. Check: your object is now 2 layers tall. Step 3: Put the yellow br on top, lined up exactly with the blue br. Check: from the front you see blue on the bottom, 2 reds, then yellow on top.

My build test

Pieces in my build	Pieces the Builder got right	First step that went wrong

Think about it

1. Why does the Builder need to know the front before Step 1?

_____

_____

2. Describer: which part of the plan was hardest to write today?

Bring to the next meeting: Put your directions behind this page in your binder and bring them to Lesson 3. The Team Captain will ask to see them.

Lab I2: Craft Words and Measuring

THIS LAB HELPS ME

- I can write using only words and numbers: no drawings, arrows, codes, or symbols, and punctuation only in its normal way.
- I can use counting and measuring to describe size and distance (bumps, holes, centimeters, halfway).
- I can describe objects made of craft materials like straws, cups, pipe cleaners, craft sticks, foam balls, and stickers.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. At the tournament the object might be made of straws, cups, and pipe cleaners. These need measurements and very exact words. In this lab you measure real objects and write directions with numbers.

Steps (30 minutes)

You need: Cup, straw, and a spoon or craft stick, Ruler with centimeters, Tape. Set the timer for each step.

1. Me 4 min. Set a timer for 4 minutes. Read Craft words out loud. Find the rim, the bottom, and the bendy part on your real cup and straw.
2. Me 6 min. Set the timer for 6 minutes. Rewrite each sentence in Fix It so it follows the rules and is clear. No letter shapes, arrows, or faces.
3. Me 6 min. Set the timer for 6 minutes. Fill in Measure it. Measure each object in centimeters, then work out where halfway and one third of the way would be. Mark halfway on your straw with a pencil.
4. Me 8 min. Set the timer for 8 minutes. Arrange the cup, straw, and spoon (or stick) in an interesting way and tape them. Write directions with at least 2 centimeter measurements. Words and numbers only.
5. Both 6 min. Set the timer for 6 minutes. Take it apart. Your Builder rebuilds it from your words. Measure how far off each piece is, in centimeters, and write it in How close?

Check your work: If a piece was off by more than 1 centimeter, find the sentence that caused it and add a better measurement. Check your Fix It answers with the Lab I2 answer card in the Open Lab bin, or at the next Huddle (different words are fine if they are clear and follow the rules).

Craft words

Straw	long or short part, bendy part, open end
Pipe cleaner	wrap (go around), twist (turn two ends together), bend in half, bend to make a square corner
Cup	rim (top edge), bottom, upright or upside down, inside, outside
Craft stick	flat side, thin edge, rounded end
Measuring	centimeters from the end, halfway, one third of the way, touching, 2 finger widths is NOT exact

Fix It: rewrite each one so it follows the rules and is clear

1. Bend the pipe cleaner into an L.

2. Put the straw -> on the cup.

3. Stick the ball on it.

4. Put stickers on the front like :)

Measure it

Object	Length (cm)	Halfway (cm from the end)	One third of the way (cm)
Straw			
Spoon or craft stick			
Cup (how tall)			

How close?

Piece	How far off (cm)	Sentence to fix

Bring to the next meeting: Bring your Measure it table and your directions to Lesson 3. You will describe a craft model that day.

Lab I3: Check Sentences and Judging

THIS LAB HELPS ME

- I can add check sentences (Now your object is 3 layers tall) so my Builder can catch mistakes.
- I can score a build like a judge: one point for each piece in the right place and connected the right way.
- I can manage time: the Describer uses all 25 minutes to write and reread, and the Builder works carefully but quickly (build time breaks ties).

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Check sentences help your Builder catch mistakes. A time plan helps you use every minute. In this lab you also score builds the way a real judge does.

Steps (29 minutes)

You need: Building pieces (6 to 8), Notebook paper, Timer. Set the timer for each step.

1. Me 3 min. Set a timer for 3 minutes. Read Tournament time plans and Good check sentences. Circle the time-plan step you usually skip.
2. Me 5 min. Set the timer for 5 minutes. Read the model in Be the judge. Write 3 check sentences for it in Write 3 checks: one about height, one about what you see from the front, and one count.
3. Me 6 min. Set the timer for 6 minutes. Score Teams A, B, and C in Be the judge. Remember: a piece counts if it is in the right place and connected right, extra pieces cost nothing, and a tie goes to the faster build. Rank the teams.
4. Me 12 min. Timed practice. Describers: build something with 6 to 8 pieces and write about it using the time plan (1 minute study, 2 overview and parts, 7 steps, 2 reread and add checks). Builders: rebuild something from written directions (a family member's, or a LEGO booklet with the pictures covered) and time yourself.
5. Me 3 min. Set the timer for 3 minutes. Use the Judge's rubric on your practice. Write your score and time on the last row.

Check your work: Check your Be the judge scores with the Lab I3 answer card in the Open Lab bin, or at the next Huddle. Describers: count your check sentences (goal: at least 3).

Tournament time plans

Describer: 25 minutes	Builder: 20 minutes
2 min: study the object from every side, set the front	2 min: read the whole description
3 min: abbreviations, overview, parts list	2 min: sort parts and match the parts list
15 min: numbered steps, bottom up	14 min: build one step at a time, check off each step
5 min: reread and add check sentences	2 min: compare to every check sentence

Good check sentences

Height	Now your object is 3 layers tall.
What you see	From the front you see red, then blue, then red.
Count	You should have used 6 pieces so far.
Leftovers	You should have 2 craft sticks left.

Be the judge: the model

6 pieces. A blue 2 by 4 brick lies flat. A red 2 by 2 brick sits on each end of it. A yellow 2 by 4 brick lies on top of both reds. A white 1 by 2 brick stands up on the front right corner of the yellow. A black 1 by 1 sits on the back left corner of the yellow.

Be the judge: score each team out of 6

Team	What the Builder built	Score	Build time
A	Everything is right, except the white 1 by 2 is on the front LEFT corner.		9 min
B	Everything is right, except the black 1 by 1 is missing. 2 extra pieces are left in the bag.		7 min
C	The white 1 by 2 is lying flat, not standing up. The black 1 by 1 is on the front left corner.		6 min
Rank	1st: ___ 2nd: ___ 3rd: ___		

Write 3 checks

1. Height check:

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2. From the front check:

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3. Count check:

Judge's rubric

Piece	Right place?	Connected right?	Point
Total and build time			

Bring to the next meeting: Bring this lab and your timed practice to Lesson 4 (the mock tournament). Decide your tournament role with your partner.